A leading Heun root at a branch point of its limiting spectral tree

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Abstract

For a third-order Heun operator with leading coefficient $z^4 - z$, we prove convergence of the complete sequence of Van Vleck root-counting measures to an explicit positive law supported on a three-spoke star. The leading root zero is a branch point, refuting the universal leaf assertion in AIM 1032. The proof uses total nonnegativity, compounds and moment determinacy on an interval.

Repository target. This manuscript concerns AIM problem 1032 at revision `3fa50884d129`. Its scope is stated below.

The universal leaf assertion in Problem 1032 is false. Consider the third-order Heun operator

$$\mathfrak{d} = (z^4 - z)\frac{d^3}{dz^3}, \qquad Q_1 = Q_2 = 0.$$

Its leading coefficient has the four simple, noncollinear roots $0, 1, \omega, \omega^2$, where $\omega = e^{2\pi i/3}$. We prove that the full sequence of Van Vleck root-counting measures converges to the explicit probability law

$$\boxed{\mu_n \Longrightarrow \mathrm{Law}(\omega^J Y X^{1/3}),} \tag{1}$$

where the independent variables satisfy

$$X \sim \mathrm{Beta}(1/3, 2/3), \qquad Y \sim \mathrm{Beta}(1, 1/3), \qquad \mathbb{P}(J = j) = \frac{1}{3} \quad (j = 0, 1, 2).$$

The support of this limit is exactly

$$T = [0, 1] \cup \omega[0, 1] \cup \omega^2[0, 1]. \tag{2}$$

Thus the leading root zero is a vertex of valency three, rather than a leaf.

This disproves the precise universal statement in the pinned catalogue and the same leaf clause in Shapiro–Tater (2008), Conjecture 3, and Shapiro (2026), Conjecture 1. The example is consistent with weak convergence, positivity, and support on a finite tree. It does not refute a conjecture restricted to generic leading polynomials; Shapiro's later Conjecture 2 explicitly has such a restriction.

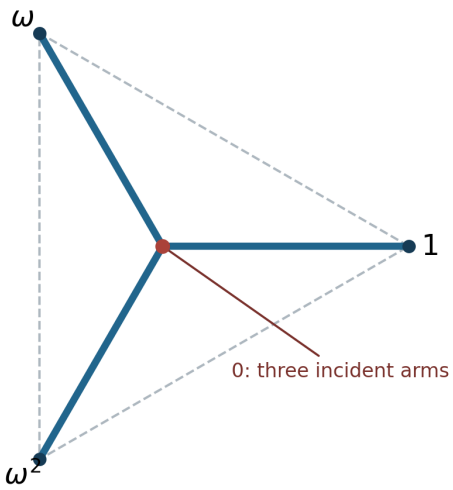


Figure 1: The exact support of the limiting root measure. The dashed triangle is the convex hull of the leading roots; the origin is a branch point with three incident arms.

1. Van Vleck roots as finite-matrix eigenvalues

For $n \geq 3$, set $\theta_n = n(n-1)(n-2)$ and define an operator on polynomials of degree at most n by

$$A_n = z - \frac{z^4 - z}{\theta_n} \frac{d^3}{dz^3}.$$

Put $c_j = j(j-1)(j-2)/\theta_n$, with $c_0 = c_1 = c_2 = 0$. On monomials,

$$A_n z^j = (1 - c_j) z^{j+1} + c_j z^{j-2}, \qquad 0 \leq j \leq n, \tag{3}$$

where a term with zero coefficient is omitted. The raising term vanishes at $j = n$, so A_n preserves this polynomial space.

If a polynomial S of degree n solves

$$(z^4 - z)S''' + V(z)S = 0,$$

comparison of the highest-degree coefficient forces $V(z) = -\theta_n(z - t)$ for some t. The equation is then $A_n S = tS$. Conversely, every eigenvector of A_n has degree n: if its highest degree were $j < n$, the nonzero raising term $(1 - c_j)z^{j+1}$ could not be cancelled. Thus the eigenvalues of A_n, counted with algebraic multiplicity, are exactly the Van Vleck roots in the problem.

If these roots are $t_{n,1}, \ldots, t_{n,n+1}$ with multiplicity, the measure under discussion is

$$\mu_n = \frac{1}{n+1} \sum_{j=1}^{n+1} \delta_{t_{n,j}}.$$

The matrix in (3) is nonnegative and each column sums to one. Its eigenvalues therefore lie in the closed unit disk. Its zeroth row vanishes. Restriction to the invariant subspace spanned by $z, z^2, \ldots, z^n$ gives an $n \times n$ matrix B_n, with

$$\det(tI - A_n) = t \det(tI - B_n). \tag{4}$$

2. Exact star support for every degree

Write $n = 3q + r$, where $r \in \{0, 1, 2\}$ and $q \geq 1$. Group the monomials $z, \ldots, z^n$ according to degrees congruent to $1, 2, 0$ modulo three, in that order. Their dimensions are

$$d_1 = q + \mathbf{1}_{\{r \geq 1\}}, \qquad d_2 = q + \mathbf{1}_{\{r=2\}}, \qquad d_3 = q.$$

In this basis,

$$B_n = \begin{pmatrix} 0 & 0 & L \\ U_1 & 0 & 0 \\ 0 & U_2 & 0 \end{pmatrix}, \tag{5}$$

where L is a $d_1 \times q$ lower bidiagonal matrix, U_1 is a $d_2 \times d_1$ upper bidiagonal matrix, and U_2 is a $q \times d_2$ upper bidiagonal matrix. In zero-based indices, their nonzero entries are

$$(U_1)_{a,a} = 1 - c_{1+3a}, \qquad (U_1)_{a,a+1} = c_{4+3a},$$

$$(U_2)_{a,a} = 1 - c_{2+3a}, \qquad (U_2)_{a,a+1} = c_{5+3a},$$

$$L_{a,a} = c_{3+3a}, \qquad L_{a+1,a} = 1 - c_{3+3a},$$

whenever the indicated row and column exist. All these entries are positive. In particular, the leading $q \times q$ blocks have positive diagonal entries and positive first off-diagonal entries in their indicated directions.

Consider the square return matrix on the third residue class,

$$C_q = U_2 U_1 L. \tag{6}$$

We next prove that its eigenvalues are positive and simple.

Total nonnegativity and the compound argument

A matrix is totally nonnegative when all minors with increasing row and column indices are nonnegative. Each of the rectangular bidiagonal factors in (6) has this property. Indeed, in any nonzero minor the bidiagonal pattern permits only an order-preserving matching of row and column indices; its determinant is therefore a product of nonnegative entries.

For $1 \leq p \leq q$, the p-th compound matrix $C_q^{[p]}$ has rows and columns indexed by the p-element subsets of $\{0, \ldots, q-1\}$, and its entries are the corresponding minors. Cauchy–Binet implies that every compound is nonnegative. Its diagonal entries are positive: in both Cauchy–Binet sums, retain the same index set in all three factors and use their positive diagonal minors.

The compound also has a positive entry whenever one occupied index is moved by one step to an adjacent vacant position, in either direction. For one direction, use the corresponding off-diagonal minor of L and diagonal minors in the other factors; for the reverse direction, use an off-diagonal minor of U_1. All intermediate choices stay within the first q indices, so the extra rows and columns in the rectangular cases cause no difficulty.

The graph of p-element subsets under these adjacent moves is connected. Thus every compound is irreducible with positive diagonal, and hence primitive. Perron–Frobenius gives it a unique simple eigenvalue of maximal modulus, which is positive. The eigenvalues of the p-th compound are the

p-fold products of eigenvalues of C_q. The compound eigenvalue identity and the Perron–Frobenius facts used here are standard matrix theory; see, for example, Gantmakher–Krein, Section 4, cited below. Successively applying this fact for $p = 1, \dots, q$ shows that the eigenvalues of C_q, in decreasing modulus order, are positive and have strictly decreasing moduli. Its determinant is positive by the principal-minor argument, so none is zero.

Write these eigenvalues as $s_1, \dots, s_q > 0$. For $t \neq 0$, block elimination in (5) gives

$$\begin{aligned} \det(tI_n - B_n) &= t^{d_1+d_2} \det(tI_q - t^{-2}C_q) \\ &= t^r \det(t^3 I_q - C_q). \end{aligned} \tag{7}$$

The identity holds for all t because both sides are polynomials. Therefore B_n has the three cube roots of each s_j and r additional zero eigenvalues. Equation (4) supplies one further zero eigenvalue for A_n.

Column stochasticity of A_n implies $0 < s_j \leq 1$. Consequently every Van Vleck root lies in the star T in (2), for every $n \geq 3$. The measure μ_n is invariant under multiplication by ω. Its mass at zero is $(r+1)/(n+1)$, which tends to zero.

3. Limiting moments from closed paths

For a fixed nonnegative integer ℓ,

$$\int z^{3\ell} \, d\mu_n(z) = \frac{1}{n+1} \operatorname{tr}(A_n^{3\ell}). \tag{8}$$

A closed path of length 3ℓ in the matrix (3) must have 2ℓ upward steps of size one and ℓ downward steps of size two. There are $\binom{3\ell}{\ell}$ orderings. For a starting index farther than 6ℓ from the endpoints, all these paths are permitted. Along each such fixed-length path,

$$c_{j+O(1)} = (j/n)^3 + O(n^{-1})$$

uniformly in the starting index j. Its weight is therefore

$$(1 - (j/n)^3)^{2\ell} (j/n)^{3\ell} + O(n^{-1}).$$

Only a bounded number of starting indices are excluded at the endpoints, and all path weights are at most one. Taking the normalized trace and using a Riemann sum gives

$$\begin{aligned} \lim_{n\to\infty} \int z^{3\ell} \, d\mu_n(z) &= \binom{3\ell}{\ell} \int_0^1 (1-t^3)^{2\ell} t^{3\ell} \, dt \\ &= \frac{\Gamma(3\ell+1)\Gamma(\ell+1/3)}{3\,\Gamma(\ell+1)\Gamma(3\ell+4/3)} =: m_\ell. \end{aligned} \tag{9}$$

The same closed-path argument gives zero for every holomorphic moment whose order is not divisible by three. Formula (9) gives $m_0 = 1$.

The exact support result from Section 2 is essential here. Holomorphic moments alone would not determine an arbitrary measure inside a two-dimensional convex hull.

4. Identification of the probability law and its full support

Let ρ_n be the pushforward of μ_n under $z \mapsto z^3$. By Section 2, ρ_n is a probability measure on $[0, 1]$. Equation (9) gives the limit of every ordinary moment.

For the independent beta variables in (1), the beta integral gives

$$\mathbb{E}[X^\ell] = \frac{\Gamma(\ell + 1/3)}{\Gamma(1/3)\Gamma(\ell + 1)},$$

$$\mathbb{E}[Y^{3\ell}] = \frac{\Gamma(3\ell + 1)\Gamma(4/3)}{\Gamma(3\ell + 4/3)}.$$

Since $\Gamma(4/3) = \Gamma(1/3)/3$, their product is exactly m_ℓ. Polynomials are dense in $C([0,1])$, so convergence of these moments proves

$$\rho_n \Longrightarrow \mathrm{Law}(XY^3). \tag{10}$$

Both beta densities are positive on $(0,1)$. Their product variable XY^3 therefore has support all of $[0,1]$.

Rotation invariance determines μ_n from ρ_n. For every continuous function f on T,

$$\int_T f(z)\, d\mu_n(z) = \int_0^1 \frac{1}{3} \sum_{j=0}^{2} f(\omega^j s^{1/3})\, d\rho_n(s).$$

The function on the right is continuous also at zero. Taking the limit in (10) proves (1), and its support is exactly the three-spoke star T.

5. Contradiction to the universal leaf clause

The roots of $Q_3(z) = z^4 - z$ are simple: $Q_3'(0) = -1$, while Q_3' equals 3 at each nonzero root. They are not all collinear. Thus the operator meets all the stated hypotheses.

Nevertheless, the support of its limiting measure contains three nontrivial arms issuing from zero. Zero has valency three and cannot be a leaf. This also contradicts the weaker interpretation that the measure need only be supported on some larger finite tree: any such closed tree contains the support T, and zero cannot become a leaf when a set containing these three arms is enlarged to another tree. Connecting two of the arms elsewhere would instead create a cycle.

A corrected statement must allow some leading roots to be internal vertices or impose an additional restriction that excludes this example. The proof supplies the actual positive limit and its full support; the counterexample is not based only on exterior moments or a numerical root plot.

6. A higher-order family

The same argument, along degrees $n = kq$, applies to

$$\mathfrak{d} = (z^{k+1} - z)D^k, \qquad k \geq 3.$$

The limit is the law of $\omega_k^J Y X^{1/k}$, where

$$\omega_k = e^{2\pi i/k}, \qquad X \sim \mathrm{Beta}(1/k, 1 - 1/k), \qquad Y \sim \mathrm{Beta}(1, 1/k),$$

and J is independently uniform on $\{0, \ldots, k-1\}$. Its support is the k-spoke star, with zero a leading root of valency k. The fully proved third-order case already suffices to disprove the universal claim.

References and verification scope

1. B. Shapiro and M. Tater, *On spectral polynomials of the Heun equation. I*, arXiv:0812.2321 (2008), Section 4, Conjecture 3, printed p. 13. https://arxiv.org/pdf/0812.2321.
2. B. Shapiro, *Van Vleck spectra of high-order Heun operators: finite-band universality and exterior asymptotics*, arXiv:2607.02700 (2026), Conjecture 1, printed p. 4. The later Conjecture 2 explicitly assumes genericity and is not refuted here. https://arxiv.org/pdf/2607.02700.
3. F. Gantmakher and M. Krein, *Sur les matrices complètement non négatives et oscillatoires*, Compositio Mathematica **4** (1937), 445–476. Section 4, p. 461, states the compound eigenvalue-products identity and the oscillatory positive-simple-spectrum theorem. https://www.numdam.org/item/CM_1937___4___445_0.pdf. The particular factorization and compound irreducibility argument are proved above.

Small finite-matrix calculations were used only as checks. Floating-point eigensolvers become unreliable for the highly nonnormal bidiagonal products at larger degrees; the exact compound argument proves the spectral location. No bibliographic-priority claim is made.